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Summary

Forward Deployed AI Engineer profile: full-stack engineer who ships production generative-AI and agentic systems end to end, from Python/FastAPI backends and LangGraph/LangChain agent flows to TypeScript/React frontends. Three years of AI and data engineering work (since November 2023) building advanced RAG platforms and MCP (Model Context Protocol) tool ecosystems adopted by real clients. Production systems, not prototypes: a criminal-background and lawsuit analysis solution that cut analysis time by up to 90% (from nearly a manual day to under 30 seconds), AI-generated assembly minutes adopted by 135 management companies, and a support knowledge base that cut response time by up to 80%. Currently studying Data Science and Artificial Intelligence at PUC Minas.

Experience

Group Software · AI Engineer

June 2025 – Present

- • Built the Group Verifica solution end to end (frontend, backend, AI, and database) for the condominium and HR markets, shipped with generative AI: under 30 seconds per criminal-background and lawsuit analysis (vs. nearly 1 manual day, ~90% reduction); 53 clients tested within 20 days; 72% demonstrated purchase interest; 196+ queries in production; R$4,750 MRR in the first 45 days
- • Architected a document-ingestion platform with a two-layer pipeline (summary + raw text), overlapping chunking, semantic search, and re-ranking: residents consult condo documents uploaded to the RAG via WhatsApp and via site/app, and support teams adopted the platform as their central knowledge base, delivering up to 80% reduction in support response time
- • Designed Group Verifica's configurable risk engine with LangGraph agent flows and tools querying real-time background-check APIs. The engine is configurable in 3 risk levels (Brando/Normal/Rigoroso) with fallback logic and full traceability of every decision, integrating legacy systems and production databases
- • Implemented AI-based transcription and automatic generation of assembly minutes and call notices: 482 meeting minutes generated in 49 days of production; 68.1% organic adoption among eligible assemblies (peak 77.3%); 1,446 specialist hours eliminated (482 × 3h per manual minute); 135 management companies adopted the feature
- • Built a smart lead-generation engine that spots cross-selling opportunities (insurance, financial, energy) from meeting transcripts: 239 leads generated in 78 days (3.1 leads/day, autonomous and continuous, no dedicated team); total AI cost of ~R$107 for the 239 leads vs. R$597–R$1,434 on market platforms (~93% savings)
- • Co-architected and developed GIA, the central AI platform that orchestrates multiple MCP (Model Context Protocol) servers: 3 Group Software products integrated simultaneously via 3 centralized MCP servers with ~20 AI tools exposed; dynamic tool routing by client entitlement and conversational context; JWT auth with a per-product flow; document chat in production via RAG over pgvector (thousands of documents indexed, adopted by internal support as the central knowledge base); recently evolved to a multi-tenant layer with per-tenant isolation and plug-and-play onboarding
- • Contributed to Group Software's multichannel conversational platform: the team delivered 6 input channels (3 WhatsApp paths, Telegram, an internal chat API, and human handoff), 21 domain entities, and 120 database migrations in ~3 months of active development (408 commits), with long-term memory between conversations, canonical identity across accounts, and a delivery pipeline to 3 AWS ECS environments provisioned via Terraform
- • Developed the production MCP tool set for the condominium-management ecosystem, consumed by the company's conversational AI platform: 19 MCP tools in production with strict Zod schemas for

structured outputs, consistent across both transports (local stdio and stateless remote HTTP); two-layer authentication (MCP client API key + end-user JWT per call); RSA-SHA256 request signing reimplemented in TypeScript with byte-for-byte compatibility with the legacy Java protocol

• Upheld the team's engineering quality bar: 2,870 automated tests across 290 files with a minimum 90% coverage gate in CI, hexagonal architecture with LangGraph/LangChain confined to the infrastructure layer, and 28 versioned formal specifications

Tech stack: Python, FastAPI, TypeScript, React, Node.js, Fastify, LangChain, LangGraph, MCP (Model Context Protocol), OpenRouter, OpenAI GPT-4o, Azure OpenAI, Claude API, pgvector, Qdrant, Redis, PostgreSQL, Azure (containers + managed PostgreSQL), AWS ECS, Terraform, Docker, OpenTelemetry, Datadog, Langfuse, promptfoo, pytest, Vitest, Zod

Use of AI:

• Builds production LLM applications daily: advanced RAG platforms (two-layer ingestion, overlapping chunking, semantic search, re-ranking) with up to 80% support response-time reduction; autonomous LangGraph agents powering a configurable 3-level risk engine with full decision traceability; AI transcription and generation of assembly minutes (482 minutes in 49 days; 1,446 specialist hours eliminated; 135 management companies); lead generation from meeting transcripts (239 leads at a total AI cost of ~R$107 vs. R$597–R$1,434 on market platforms)

• Runs multi-model orchestration via OpenRouter with systematic model comparison (GPT-5-mini, GPT-4.1-mini, Claude 3 Haiku, Gemini, Gemma) and per-conversation/client/model LLM cost auditing

• Treats LLM safety and operations as first-class: automated red teaming with promptfoo (client isolation, auth bypass, instruction injection) plus LLM observability via OpenTelemetry and Datadog, with self-hosted Langfuse

• Operates MCP servers that expose product APIs as AI-consumable tools (19 production tools with strict Zod structured-output schemas), plus the GIA multi-MCP orchestration layer routing ~20 tools by entitlement and conversational context

• Uses agentic coding tools daily (Claude Code, Cursor, GitHub Copilot, Codex) across design, implementation, testing, and review

H&P · Data Scientist / AI Engineer

November 2023 – June 2025

• Redesigned the relational database structure with software-engineering best practices: 90 tables redesigned with a 70% performance increase and an estimated ~60% saving in database-maintenance time, preparing the infrastructure to support AI workflows

• Pioneered the deployment of generative-AI solutions across the company, from identified strategic use cases, delivering up to 80% reduction in report-writing time for the responsible team

• Ran internal AI study groups on RAG, chunks, embeddings, and LLM selection: 12 professionals upskilled during the program

• Owned governance and integrity of the Fundão dam breach (Mariana/MG) registration databases: absolute data accuracy for strategic decisions and high-impact reports, serving 1,200 people

Tech stack: Relational database architecture and governance; generative-AI (LLM) solutions; RAG, chunking, embeddings, and LLM selection (internal AI study program)

Use of AI: Pioneered company-wide generative-AI adoption at H&P: shipped LLM solutions that automated report writing (up to 80% time reduction for the responsible team) and built internal AI capability by running study groups on RAG, chunking, embeddings, and LLM selection (12 professionals upskilled).

The Brooklyn Brothers · Mid-Level Front-End Developer

March 2022 – January 2023

• Developed and improved high-performance web interfaces for Rexona, one of the top clients in the global portfolio: 23% SEO improvement plus better user experience, operational efficiency, and accessibility; ~30% performance improvement

• Worked hand-in-hand with development teams in India and the United Kingdom in a global team of 15 people, in a multilingual, agile (Scrum) environment

Tech stack: React, Context API, JavaScript (ES6+), HTML5, CSS3, responsive web components

Use of AI: AI tooling was not part of this role, which predates my AI engineering work (2022); my AI track record is documented at Group Software and H&P.

Kenzie Academy Brasil · Peer Coach (Teaching Assistant)

August 2021 – February 2022

- Coached 100 students on React, CSS, and HTML (Virtual DOM, styled-components) and versioning with Git/GitHub, who raised the quality of their projects
- Clarified Context API, Redux, API consumption, and unit-testing questions, enabling more effective problem-solving among students

Tech stack: React, CSS, HTML, Git/GitHub, Context API, Redux

Use of AI: AI tooling was not part of this role; my AI track record is documented at Group Software and H&P.

Red Lion Inglês Britânico · English Instructor

March 2020 – May 2020

- Taught English to students of all age groups, creating didactic materials and interactive activities and tracking progress with tests and constructive feedback

Tech stack: None (classroom teaching role)

Use of AI: Not applicable to this teaching role; my AI track record is documented at Group Software and H&P.

Education

Web Development · Kenzie Academy Brazil

May 2021 – May 2022

Core Skills

- **GenAI (Generative AI) and LLMs:** OpenAI, OpenRouter, Azure OpenAI, Claude API, GPT, Gemini, Anthropic, prompt engineering, LLM deployment, speech-to-text, text-to-speech, voice AI
- **Agentic AI, Agents and RAG:** AI agents, agentic AI, LangChain, LangGraph, advanced RAG, multi-agent systems, agent tool integration, tool/function calling, MCP (Model Context Protocol), semantic search, context engineering, embeddings, vector search, vector databases (pgvector, Qdrant, Chroma, Pinecone)
- **AI Tooling and Ops:** Claude Code, Cursor, GitHub Copilot, Codex, AI evaluations, LLMOps practices (automated red teaming, LLM observability, per-conversation LLM cost auditing), AI observability, CI/CD, Docker, Terraform, Azure, Azure AI Foundry, Vertex AI, pytest
- **Languages and Frameworks:** Python, TypeScript, JavaScript, FastAPI, React, SQLAlchemy, Node.js, Vitest
- **Data and Machine Learning:** PostgreSQL, MongoDB, Redis, data engineering, API development, pandas, Power BI, Machine Learning fundamentals (LLM evaluation and model selection)
- **Ways of Working:** Agile methodologies, Scrum, automated testing, multi-tenancy, hexagonal architecture

Certifications

Model Context Protocol (MCP) Professional Certification · Anthropic (certificate of completion), 2026

Model Context Protocol: Advanced Topics · Anthropic (certificate of completion)

Introduction to Model Context Protocol · Anthropic (certificate of completion)

Introduction to Subagents · Anthropic (certificate of completion)

Introduction to Agent Skills · Anthropic (certificate of completion)

Languages

Portuguese (native)
English (full professional)

Projects

Personal projects

Nexus Agent

WhatsApp AI agent for appointment scheduling in the health and beauty sectors (AI Appointment Setter), live at AgendaFacilitada.com on a self-hosted VPS; integrates LLMs (LangChain/LangGraph) with calendar APIs and natural-language processing for fluid WhatsApp conversations (Evolution API).

Python · LangChain · LangGraph · WhatsApp Business API · Google Calendar API · Evolution API

Financeiro

Personal finance and investment platform in daily use by my family, in production on a self-hosted VPS (Docker Compose + Caddy + PostgreSQL with automated backups; 1.3k+ commits since 2026-08); exposes 35 MCP tools in production (bearer-authenticated) so AI clients operate the finances in natural language; 667 automated tests (pytest/asyncio) covering finance rules, authentication, and MCP tools.

Python · FastAPI · NiceGUI · SQLAlchemy · PostgreSQL · MCP (Model Context Protocol) · Docker Compose · Caddy · pytest

Termômetro da Massa

PUC Minas academic project (text-mining coursework): an ABSA (aspect-based sentiment analysis) pipeline measuring Atlético Mineiro fans' perception of each player from public X (Twitter) and Instagram comments; raw, silver, and gold data layers with dedup; LLM-based aspect extraction (OpenAI); gold dataset with inter-annotator kappa (Streamlit annotation app); 453 automated tests. This project is my machine-learning fundamentals and evaluation evidence.

Python · Apify · OpenAI · pandas · scikit-learn · PyArrow (Parquet) · Streamlit · pytest